PHASE 2 EXPERIMENTAL RESULTS

Cost-Aware Allocation for Performance-Maximizing Explainable Ensembles

1. Research question and evidence chain

Phase 2 investigates three distinct limitations of the original EEG framework: routing instability under perturbations, the opacity of the allocator itself, and unnecessary pre-routing computation.

Identified gap → proposed solution → measurable prediction

Gap 3	Small feature perturbations can flip model assignments and explanations.
Gap 6	The allocator is opaque; users cannot see why it routed an observation.
Cost-aware	The original gate needs black-box outputs before routing, wasting computation.
Solutions	SR-EEG stabilizes routing; TreeSHAP explains routing; Cost-Aware EEG enables conditional execution.
Evidence	Separate results are reported for the 24-dataset stability, SHAP, and Wine cost benchmarks.

Experimental design

Gap 3 and Gap 6 use 24 datasets, seeds {0, 1, 2}, five perturbation levels, and q values from 0.1 to 0.9. The cost-aware experiment uses Wine (N=537), $q \in \{0.20, 0.40, 0.60, 0.80\}$, validation-only thresholds, and 50 timing repetitions after 10 warm-up runs.

Reporting integrity

Published values are labeled Published. Local baseline values are labeled Reproduced Original EEG. Only Cost-Aware EEG values are presented as Phase-2 results.

PUBLISHED EVIDENCE FROM THE ORIGINAL PAPER

Extracts are reproduced from the supplied XAI Paper.pdf; they are not our experimental results

Extract A — Results discussion and metric definitions (original paper printed p. 14621; PDF p. 5)

Metrics

We define the following metrics which are used to measure performance of our method. First we define the Percentage Performance Captured over Random (PPCR) for a given allocator as follows: $PPCR(a_q) = \frac{AUC(a_q) - AUC(r_q)}{AUC(o_q) - AUC(r_q)}$ where $AUC(f_q)$ is the area under the curve of function f_q over all values of q in its domain, o_q is the oracle allocator which has perfect information on the whole dataset, and r_q is the random allocator which selects a subset from the data being allocated uniformly at random. The PPCR metric is a percentage and represents the proportion of the oracle AUC, in excess of that also covered by random allocation, that the learned allocator is able to capture. Thus a value of zero indicates performance on par with r_q and a value of one represents perfect allocation.

Next, we define the Percent Q Equal or Over Max (PQEOM) as the percentage of q values for which the allocator is performing at least as well as the most accurate ensemble member model (i.e. g or b) and Percent Q Over Max (PQOM) as the percentage of q values for which the allocator is performing better than the most accurate ensemble member model (i.e. g or b).

Next, we define the Percent Contribution of Feature-dependent Allocator (PCFA) as the percentage of q values for which the feature dependent allocator a'_q is used for allocation decisions as opposed to the feature-independent allocator a''_q. A value close to one indicates that a'_q is used often, while a value close to zero indicates it is a''_q instead.

Next, we define the 95% Threshold Q Max (95TQM) as the highest value of q for which the ensemble performance meets or exceeds 95% of the performance of the better of g and b. Thus this is a measure of how much explainability can be utilized before the performance price becomes material.

Next, we define the maximum accuracy achieved by the allocator across all q (Max Acc), and the highest value of q for which this accuracy is maintained (Argmax q). The Max Acc can be benchmarked against the AUC, interpretable as the average accuracy across q. Each of these metrics is a percentage and higher values correspond with higher performance and higher explainability at this maximum performance level, respectively.

Finally, we define the accuracy with which the four sufficiency categories (Z_y, Z_1, Z_2, and Z_0) can be estimated as the sufficiency accuracy (s Acc). The higher this accuracy, the better able the allocator is to inform the user of which category a given observation is likely to be a member of.

Results

Evaluation of allocator performance using the metrics defined previously as well as visual inspection of the performance vs explainability trade-off curves (Fig. 2) revealed both the benefits and some of the limitations of learned allocation in the tabular data setting.

Firstly, performance was found to consistently and significantly outperform random allocation, as quantified by a cross-dataset PPCR of 37% (Table 2), indicating that the learned allocation captured close to 40% of the area under the curve available and in excess of random allocation. It was also found that on some datasets in particular, learned allocation performed close to oracle allocation (e.g. 89% and 71% on the IsoletR and BrazilianHousesR datasets).

Learned allocation was also found to perform at least at the level of the best ensemble member model across an average of 74% of the explainability range (PQEOM in Table 2). This indicates that for many datasets, there is a substantial explainability "free lunch" to be taken advantage of without performance loss. On a few datasets, performance of the allocated ensemble was found to outperform both g and b for a majority (93%) of the q range (PolR and FifaR PQOM). The 95TQM metric also supported these conclusions, with a cross-dataset average value of 94% indicating that allocation performance was within 5% of maximal individual model performance across approximately all values of q.

Assessing the PCFA metric suggests some limits to the upside of learned, feature dependent allocation - at least in the tested tabular data setting. A cross-dataset average value of $35\% \pm 34\%$ indicates that on average, the range for which the feature dependent allocator is used over the feature-independent one is indistinguishable from zero. This is consistent with a visual inspection of the representative performance-explainability curve e.g. Fig. 2 (b) where there is no improvement to be had in excess of a''_q. However, it is noted that the only possibility for "homerun" allocations is through the feature dependent a'_q as seen in Fig 2 (a) with the PolR dataset and also in Table 2 for datasets SulfurR, BikeSharingR, and FifaR. Thus the ensembled allocation scheme offers this upside without downside risk of low performance in either a'_q or a''_q.

Evaluation of the case in which a single allocation is needed is also positive. On a cross-dataset average, the 84% maximal accuracy achieved is quite high, and is also achieved at a high average explainability level (64%). Particularly strong individual results can be seen in the Pol and SulfurR datasets (Table 2). We also find that on a observation-level, the allocation is an accurate estimator of sufficiency category, with a cross-dataset average of 76% and with few datasets with accuracy under 60%.

Ablation Studies

Allocator Feature Set Selection In addition to the features x used to learn the glass box and black box models, the allocation task also has access to their predictions $g(x)$ and $b(x)$, and any functions of the two - since the allocator is learned subsequent to the training of these models. To obtain the optimal feature set for allocation, standard tuning procedures (e.g. cross validation) can be employed to evaluate all feature sets of interest. However, as each candidate feature set requires the training of a corresponding allocator for evaluation, this approach can be prohibitively costly.

Thus, the following study was conducted to determine whether a consistently best feature set exists for the tabular data context used in the experiments. First, the universe of candidate features was selected to be the original features x used as inputs for the ensemble component models, the predictions of both of these models $g(x)$ and $b(x)$, and finally two measures of discrepancy between the predictions, the cross-entropy $d_{ce}(g(x), b(x))$ and the mean squared error $d_{mse}(g(x), b(x))$. The measures of disagreement were included as features as they translate to the "feature independent" strategy of allocation to model b for low values of $\hat{d}(a(x), b(x))$ - in other words the optimal allocation strategy assuming a is always correct.

The candidate features were grouped into the sets listed in Table 3 and then used to train allocators on a subset of the benchmark datasets (Wine, WineR, Phoneme, SulfurR, Bank, BrazilianHousesR, FifaR, KDDIPUMS) with 6 replicates per model.

Next each feature set was evaluated as follows. First, within each dataset, each feature set's performance (defined as the AUC) was compared to the performance of the best alternative set of features. Then, the proportion of datasets for which the feature set being evaluated was not significantly worse (i.e. either significantly better or not significantly different) than the best alternative was recorded and reported in Table 3 for three significance levels (10%, 5%, 1%).

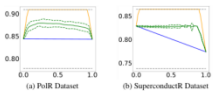
The results support the following three conclusions. First, no one feature set proved universally best across the tested datasets and thus a full search across feature sets would be advised in settings without resource constraint. Second,

The source text reports cross-dataset PPCR = 37%, PQEOM = 74%, 95TQM = 94%, maximum accuracy = 84%, and sufficiency accu

Extract B — Original paper Table 2 (original paper printed p. 14622; PDF p. 6)

Dataset	AUC	PPCR	PQEOM	PQOM	PCFA	95TQM	Max Acc	Argmax q	s Acc
Wine	79 ± 0	21 ± 0	71 ± 0	0 ± 0	7 ± 0	98 ± 0	80 ± 0	70 ± 0	78 ± 0
Phoneme	87 ± 0	12 ± 1	78 ± 4	7 ± 2	6 ± 1	100 ± 0	87 ± 0	50 ± 34	81 ± 2
KDDIPUMS	88 ± 0	17 ± 1	65 ± 3	34 ± 5	17 ± 7	100 ± 0	88 ± 0	66 ± 9	80 ± 1
EyeMovements	66 ± 0	33 ± 0	55 ± 1	7 ± 6	15 ± 2	70 ± 0	68 ± 0	31 ± 24	52 ± 0
Pol	98 ± 0	49 ± 0	98 ± 0	2 ± 0	0 ± 0	100 ± 0	99 ± 0	98 ± 0	96 ± 0
Bank	76 ± 0	-19 ± 0	4 ± 1	0 ± 0	0 ± 0	100 ± 0	79 ± 0	100 ± 0	71 ± 0
MagicTelescope	86 ± 0	39 ± 0	87 ± 2	12 ± 11	10 ± 0	100 ± 0	86 ± 0	47 ± 28	82 ± 1
House16H	89 ± 0	40 ± 0	84 ± 4	9 ± 6	6 ± 2	98 ± 0	89 ± 0	82 ± 8	86 ± 0
Credit	78 ± 0	5 ± 1	56 ± 4	14 ± 19	95 ± 0	100 ± 0	78 ± 0	76 ± 6	72 ± 0
California	90 ± 0	52 ± 0	88 ± 0	0 ± 0	7 ± 0	98 ± 0	91 ± 0	88 ± 0	90 ± 0
Electricity	92 ± 0	58 ± 0	88 ± 0	0 ± 0	7 ± 0	98 ± 0	93 ± 0	88 ± 0	92 ± 0
Jannis	79 ± 0	30 ± 0	53 ± 5	21 ± 5	14 ± 1	98 ± 0	79 ± 0	35 ± 6	76 ± 0
MiniBooNE	94 ± 0	54 ± 0	90 ± 0	0 ± 0	5 ± 0	100 ± 0	94 ± 0	90 ± 0	92 ± 0
WineR	73 ± 0	43 ± 0	85 ± 0	10 ± 0	20 ± 0	90 ± 0	74 ± 0	78 ± 0	67 ± 0
IsoletR	91 ± 0	89 ± 0	68 ± 0	0 ± 0	49 ± 10	73 ± 0	95 ± 0	68 ± 0	87 ± 1
CPUR	75 ± 0	40 ± 2	52 ± 18	0 ± 0	29 ± 11	80 ± 0	77 ± 0	70 ± 0	59 ± 1
SulfurR	98 ± 0	63 ± 2	73 ± 9	1 ± 2	79 ± 4	100 ± 0	98 ± 0	84 ± 22	96 ± 0
BrazilianHousesR	96 ± 0	71 ± 1	83 ± 0	64 ± 6	1 ± 2	93 ± 0	98 ± 0	8 ± 3	88 ± 0
AileronR	75 ± 0	13 ± 0	70 ± 10	5 ± 7	4 ± 2	100 ± 0	76 ± 0	40 ± 35	64 ± 0
MiamiHousingR	76 ± 0	44 ± 0	76 ± 0	0 ± 0	67 ± 10	80 ± 0	78 ± 0	75 ± 0	65 ± 0
PolR	88 ± 0	44 ± 1	96 ± 1	93 ± 1	88 ± 0	100 ± 0	88 ± 0	81 ± 2	84 ± 0
ElevatorsR	75 ± 0	25 ± 0	64 ± 4	2 ± 2	17 ± 0	88 ± 0	75 ± 0	51 ± 29	63 ± 0
BikeSharingR	77 ± 0	26 ± 1	88 ± 4	22 ± 16	82 ± 1	100 ± 0	78 ± 0	21 ± 13	72 ± 0
FifaR	77 ± 0	33 ± 0	95 ± 0	93 ± 0	78 ± 0	100 ± 0	77 ± 0	50 ± 0	69 ± 0
CaliforniaR	78 ± 0	38 ± 0	73 ± 0	68 ± 0	68 ± 0	93 ± 0	79 ± 0	42 ± 0	63 ± 0
HousesR	78 ± 0	41 ± 0	73 ± 0	0 ± 0	48 ± 1	80 ± 0	79 ± 0	73 ± 0	62 ± 0
SuperconductR	83 ± 0	42 ± 0	60 ± 13	24 ± 11	95 ± 0	95 ± 0	83 ± 0	57 ± 1	76 ± 0
HouseSalesR	76 ± 0	50 ± 1	64 ± 12	0 ± 0	56 ± 6	85 ± 0	78 ± 0	78 ± 0	67 ± 0
House16HR	92 ± 0	55 ± 0	90 ± 0	0 ± 0	6 ± 0	98 ± 0	92 ± 0	90 ± 0	86 ± 0
DiamondsR	70 ± 0	19 ± 0	83 ± 0	34 ± 6	73 ± 0	100 ± 0	71 ± 0	20 ± 6	65 ± 0
MedicalChargesR	86 ± 0	24 ± 0	91 ± 1	85 ± 1	0 ± 0	100 ± 0	86 ± 0	67 ± 2	83 ± 0
Average	**83 ± 9**	**37 ± 21**	**74 ± 19**	**20 ± 29**	**35 ± 34**	**94 ± 9**	**84 ± 8**	**64 ± 24**	**76 ± 12**

Table 2: This table summarizes allocation performance across several metrics on each dataset and, in the bottom row, across the datasets (all metrics are reported as percentages). Averages and standard deviations are reported over 5 replicates. Metric definitions can be found in the Metrics section, and discussion of results can be found in the Results section.



(a) PolR Dataset (b) SuperconductR Dataset

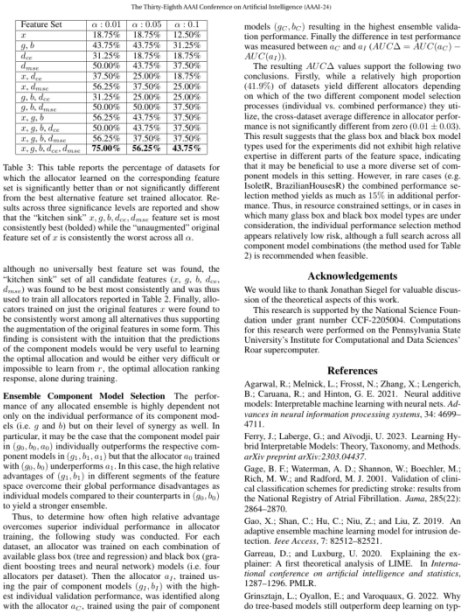
Figure 2: This figure shows two examples of the explainability (x-axis) vs sufficient performance (y-axis) trade-off, comparing the random (blue), oracle (orange), and learned (green) allocation curves. The PolR dataset is an example of complementary g and b models, resulting in an allocated ensemble that outperforms both component models across most of the q range. The SuperconductR dataset is an example of an explainability "free lunch" in which the b accuracy is maintained while increasing explainability using the allocator. Curves for all datasets are available in the Appendix of the long form paper available on arxiv.org.

The Wine row in the published table reports AUC 79, PPCR 21, PQEOM 71, 95TQM 98, Max Acc 80, Argmax q 70, and s Acc 78 (all percent

PUBLISHED FEATURE-SET RESULT AND DESIGN RATIONALE

Original paper printed p. 14623 (PDF p. 7) is used as supporting evidence

Extract C — Feature-set ablation (original paper printed p. 14623; PDF p. 7)



Feature Set	$\alpha : 0.01$	$\alpha : 0.05$	$\alpha : 0.1$
x	18.75%	18.75%	12.50%
g, b	43.75%	43.75%	31.25%
d_{ce}	31.25%	18.75%	18.75%
d_{mse}	50.00%	43.75%	37.50%
x, d_{ce}	37.50%	25.00%	18.75%
x, d_{mse}	56.25%	37.50%	25.00%
g, b, d_{ce}	31.25%	25.00%	25.00%
g, b, d_{mse}	50.00%	50.00%	37.50%
x, g, b	56.25%	43.75%	37.50%
x, g, b, d_{ce}	50.00%	43.75%	37.50%
x, g, b, d_{mse}	56.25%	37.50%	37.50%
x, g, b, d_{ce}, d_{mse}	**75.00%**	**56.25%**	**43.75%**

Table 3: This table reports the percentage of datasets for which the allocator learned on the corresponding feature set is significantly better than or not significantly different from the best alternative feature set trained allocator. Results across three significance levels are reported and show that the "kitchen sink" x, g, b, d_{ce}, d_{mse} feature set is most consistently best (bolded) while the "unaugmented" original feature set of x is consistently the worst across all α.

although no universally best feature set was found, the "kitchen sink" set of all candidate features (x, g, b, d_{ce}, d_{mse}) was found to be best most consistently and was thus used to train all allocators reported in Table 2. Finally, allocators trained on just the original features x were found to be consistently worst among all alternatives thus supporting the augmentation of the original features in some form. This finding is consistent with the intuition that the predictions of the component models would be very useful to learning the optimal allocation and would be either very difficult or impossible to learn from x, the optimal allocation ranking response, alone during training.

Ensemble Component Model Selection The performance of any allocated ensemble is highly dependent not only on the individual performance of its component models (i.e. g and b) but on their level of synergy as well. In particular, it may be the case that the component model pair in (g_0, b_0, a_0) individually outperforms the respective component models in (g_1, b_1, a_1) but that the allocator a_0 trained with (g_0, b_0) underperforms a_1. In this case, the high relative advantages of (g_1, b_1) in different segments of the feature space overcome their global performance disadvantages as individual models compared to their counterparts in (g_0, b_0) to yield a stronger ensemble.

Thus, to determine how often high relative advantage overcomes superior individual performance in allocator training, the following study was conducted. For each dataset, an allocator was trained on each combination of available glass box (tree and regression) and black box (gradient boosting trees and neural network) models (i.e. four allocators per dataset). Then the allocator a_f, trained using the pair of component models (g_f, b_f) with the highest individual validation performance, was identified along with the allocator a_C, trained using the pair of component models (g_C, b_C) resulting in the highest ensemble validation performance. Finally the difference in test performance was measured between a_C and a_f ($AUC\Delta = AUC(a_C) - AUC(a_f)$).

The resulting $AUC\Delta$ values support the following two conclusions. Firstly, while a relatively high proportion (41.9%) of datasets yield different allocators depending on which of the two different component model selection processes (individual vs. combined performance) they utilize, the cross-dataset average difference in allocator performance is not significantly different from zero (0.01 ± 0.03). This result suggests that the glass box and black box model types used for the experiments did not exhibit high relative expertise in different parts of the feature space, indicating that it may be beneficial to use a more diverse set of component models in this setting. However, in rare cases (e.g. IsoletR, BrazilianHousesR) the combined performance selection method yields as much as 15% in additional performance. Thus, in resource constrained settings, or in cases in which many glass box and black box model types are under consideration, the individual performance selection method appears relatively low risk, although a full search across all component model combinations (the method used for Table 2) is recommended when feasible.

Acknowledgements

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The paper reports that the all-feature set x, g, b, dce, dmse was most consistently strong, while the unaugmented x-only allocator was consistently worst in its benchmark. This creates the deployment tension addressed in Phase 2: the strongest allocation features include model outputs, but those outputs require model execution before routing.

How Phase 2 uses this evidence

The proposed change does not claim that X-only allocation is universally more accurate. It changes the deployment objective by removing pre-routing black-box execution. The experiments therefore measure the explicit trade-off between computational savings and predictive performance, rather than hiding the information loss.

Important distinction

Pages 14621–14623 are source-paper extracts. They document the published EEG benchmark and motivate the gap; they are not evidence generated by this Phase-2 implementation.

GAP 3 — PERTURBATION STABILITY

Original EEG vs. Stability-Robust EEG (SR-EEG), 24 datasets

Research gap and proposed improvement

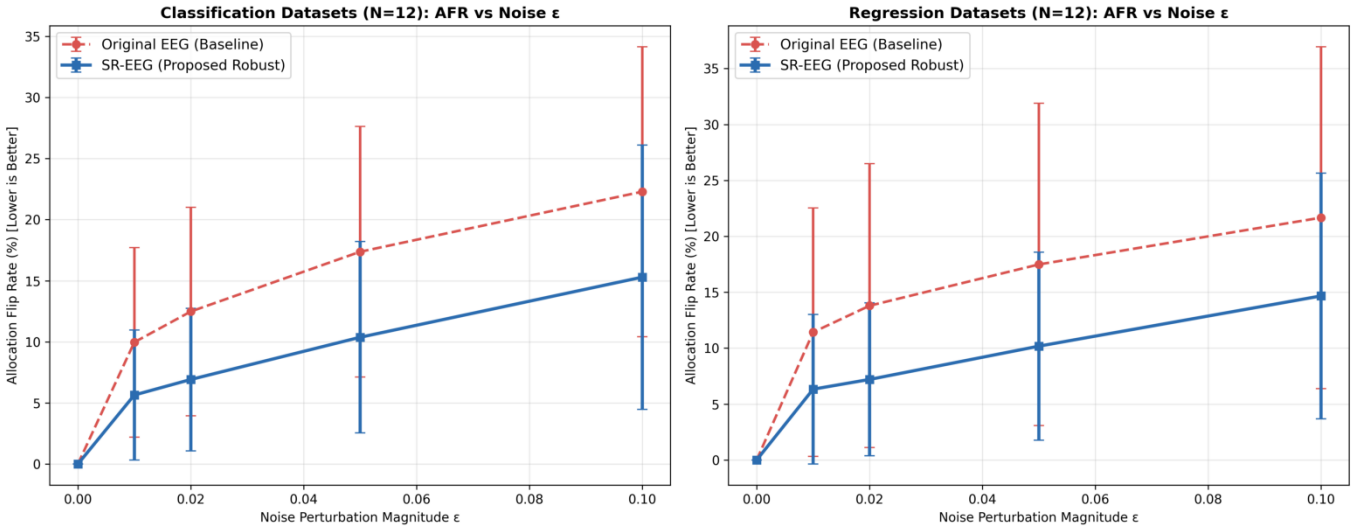
The original EEG allocator is trained on clean rankings and then applies a hard threshold. Small normalized feature perturbations can move an observation across the routing boundary. SR-EEG adds neighborhood perturbation augmentation and regularization so nearby inputs receive more consistent allocation decisions.

Table 3. Cross-dataset mean at ε=0.05 (24 datasets, all q and seeds).

Method	AFR (lower)	Prediction consistency (higher)	Clean performance	Perturbed performance
Original EEG	17.43%	41.46%	41.59%	37.73%
SR-EEG (Proposed)	10.28%	41.85%	41.45%	37.89%

Representative benchmark findings

At ε=0.05, documented reductions include CPU_R: 27.74% → 7.77% (72.0%), California: 21.84% → 9.67% (55.7%), BrazilianHouses_R: 40.47% → 18.66% (53.9%), and Wine: 14.68% → 8.56% (41.7%). These values are from results/stability_master.csv and PHASE2_RESULTS_GAP3.md.



Gap 3 conclusion: SR-EEG improves routing stability while preserving the task objective; it addresses robustness, not inference cost.

GAP 6 — META-EXPLAINABILITY

Explaining the EEG allocator with TreeSHAP

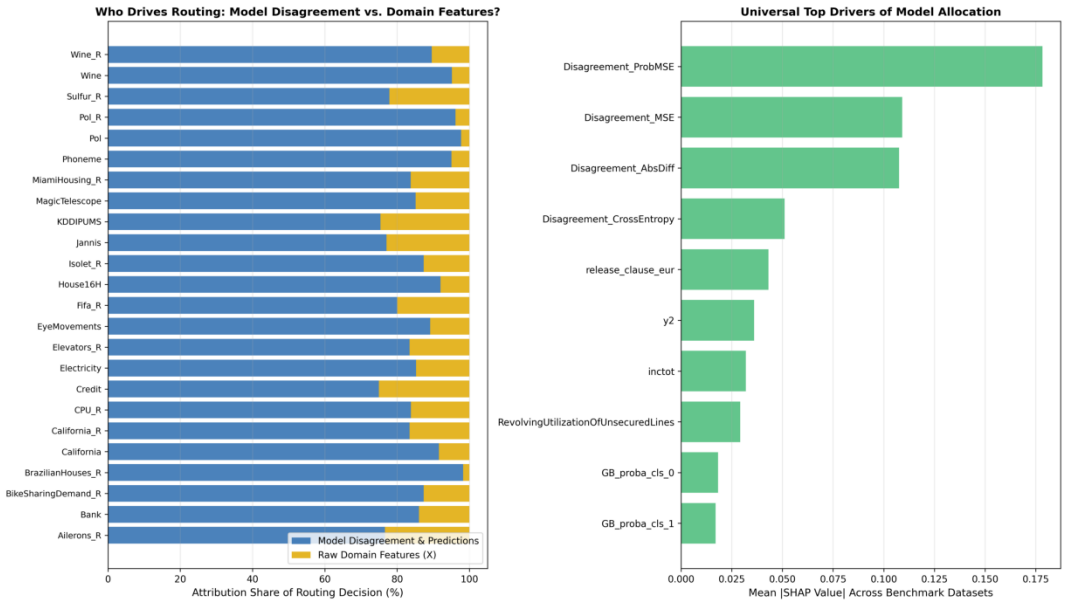
Research gap and proposed improvement

EEG routes observations for explainability, but its gradient-boosted allocator is itself opaque. Gap 6 asks: who explains the explainer? TreeSHAP quantifies whether routing is driven by raw domain features or by disagreement and confidence between component models.

Table 4. Mean TreeSHAP attribution shares across 24 datasets.

Allocator	Disagreement + outputs	Raw domain features	Interpretation
Original EEG	86.34%	13.66%	Disagreement/model confidence
SR-EEG	77.98%	22.02%	Disagreement remains dominant; shift is dataset-dependen

The SHAP summary shows that original EEG routing is primarily driven by component disagreement and predictive confidence rather than domain features. This makes individual routing decisions auditable: users can inspect whether an observation was sent to the black-box because the two component models disagreed.



Gap 6 conclusion: TreeSHAP makes the routing mechanism inspectable; it addresses meta-explainability, not perturbation stability or c

BEFORE-AND-AFTER RESULTS

Published EEG values, local Original_EEG reproduction, and Phase-2 CostAware_EEG

Published Wine metric	Value	Source/status
AUC	79	Published Table 2, p. 14622
PPCR	21%	Published Table 2, p. 14622
PQEOM	71%	Published Table 2, p. 14622
95TQM	98%	Published Table 2, p. 14622
Max Acc	80%	Published Table 2, p. 14622
Argmax q	70%	Published Table 2, p. 14622
Sufficiency Acc	78%	Published Table 2, p. 14622

Table 1. Published source-paper values (not generated by this project).

Table 2. Local experimental comparison (mean ± standard deviation across seeds [0, 1, 2]).

Quota	Original_EEG accuracy	CostAware_EEG accuracy	Latency reduction	Cost saved
q=0.20	81.38 ± 0.32%	81.13 ± 0.43%	27.52 ± 1.11%	27.93 ± 0.32%
q=0.40	80.94 ± 0.43%	80.45 ± 0.37%	38.48 ± 1.21%	44.03 ± 0.32%
q=0.60	80.45 ± 0.37%	80.82 ± 0.67%	50.32 ± 1.92%	60.59 ± 0.93%
q=0.80	80.57 ± 0.39%	78.27 ± 0.86%	60.22 ± 1.51%	73.54 ± 0.40%

The original implementation results above are obtained from baseline_results.csv using the Original_EEG path. The proposed results are obtained from cost_aware_results.csv using the matched_coverage CostAware_EEG path. The published paper values are not substituted for either local experiment.

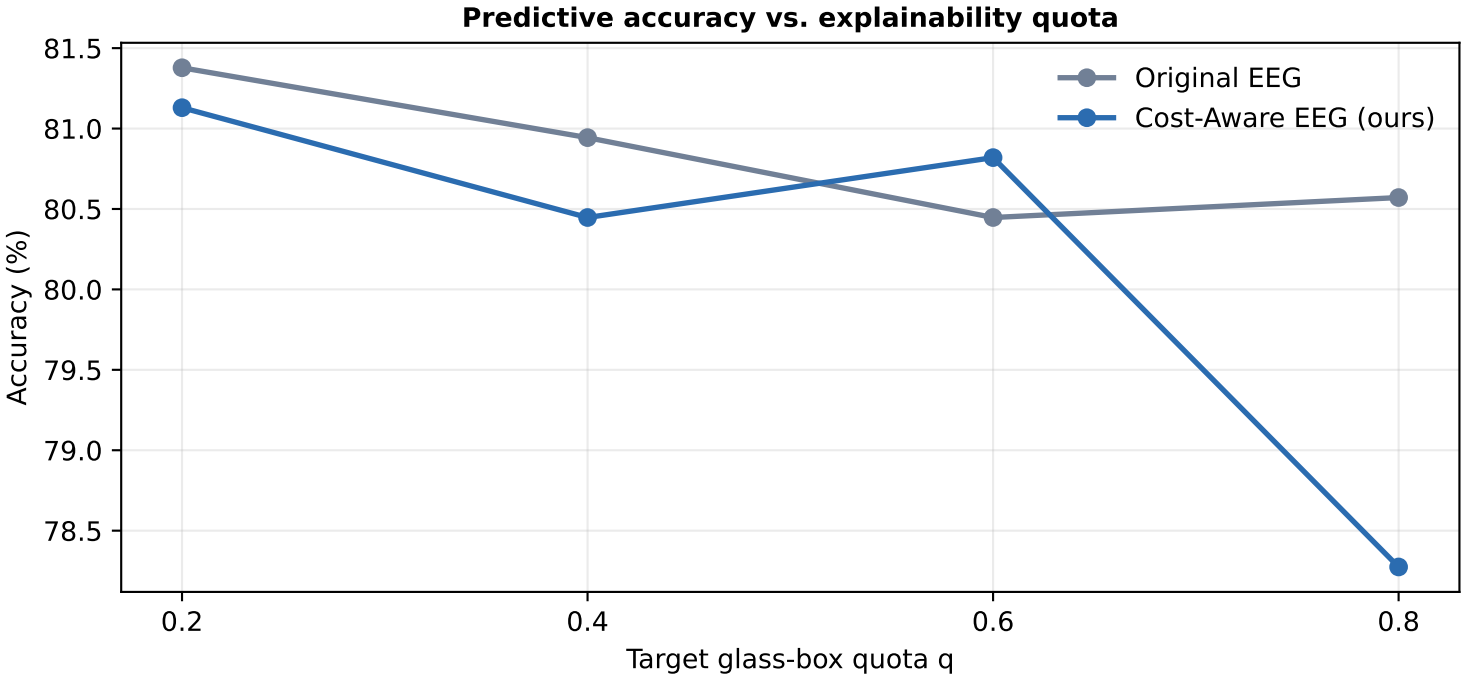
Component timing evidence

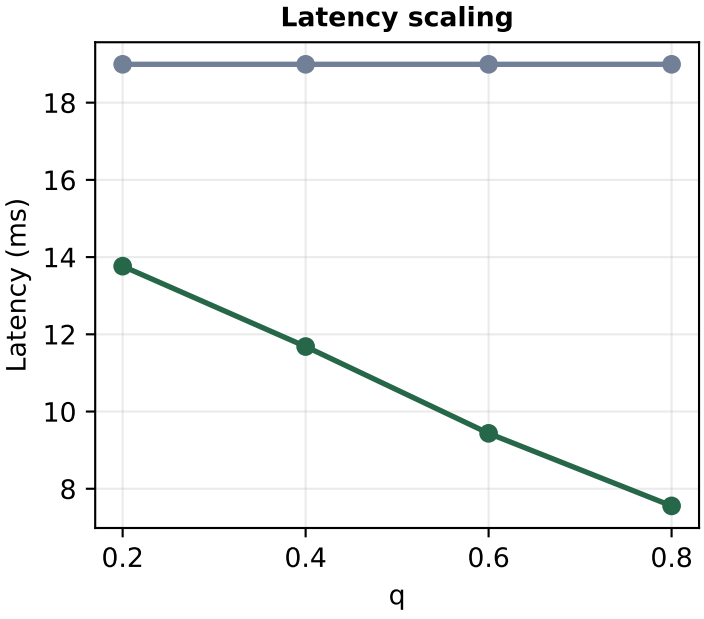
Measured component	Mean latency (N=537)
1 GlassBox Inference	0.046 ms
2 BlackBox Inference	11.341 ms

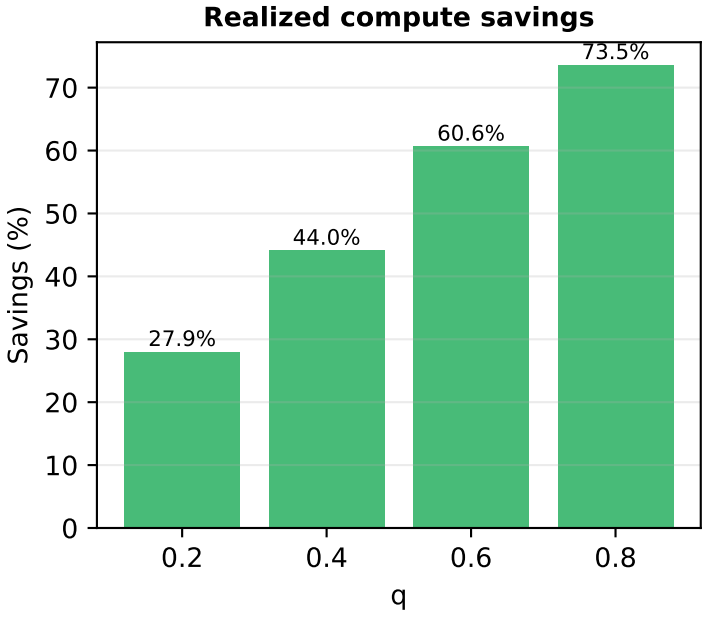
The black-box component is approximately 241× slower than the glass-box component in the stored benchmark timing. This asymmetry explains why conditional execution is a meaningful computational intervention.

VISUAL BEFORE-AND-AFTER COMPARISON

All plotted values are computed from Phase-2 result CSVs







Interpretation: the proposed method produces a monotonic latency and compute benefit as q increases, while the original pipeline remains approximately flat because the black-box model is evaluated before routing. The accuracy cost is small at moderate quotas and larger at q=0.80.

Figure note: these are implementation results, not screenshots or re-plots of the source paper.

INTERPRETATION, SUPPORTING LITERATURE, AND REPRODUCIBILITY

Connection to the proposed improvement

BranchyNet supports the central design principle used here: easy cases should exit through a cheaper computation path, while difficult cases receive more expensive processing. Cost-Aware EEG adapts that principle to a tabular explainable ensemble. Unlike BranchyNet's neural early exits, this implementation routes between a glass-box and black-box model and retains an explicit explainability quota.

Main findings

- Gap 3: SR-EEG reduces allocation flips under controlled perturbations across the 24-dataset benchmark.
- Gap 6: TreeSHAP shows that routing is primarily driven by component disagreement and confidence, making the allocator auditable.
- At q=0.20–0.60, the proposed method remains close to the reproduced original accuracy while reducing measured latency and realized cost.
- At q=0.80, the accuracy decrease is reported rather than hidden; it is the observed price of removing model-output features from the pre-routing gate.
- The three extensions address different failure modes: robustness, auditability, and deployment cost.

References

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Reproducibility record

Dataset: Wine / OpenML benchmark; test batch N=537; seeds 0, 1, 2; validation-only threshold selection; timing repetitions: 50 with 10 warm-ups. Source files: experiment_config.json, baseline_results.csv, cost_aware_results.csv, comparison_results.csv, inference_timing.csv, and the Phase-2 implementation scripts. Generated report: phase2_experimental_results_comparison.pdf.